High-Speed Rail Vehicle Design for Increased Curve-Negotiation Speed

Lateral frame shifting and differential tilt for active mass placement and carbody orientation

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Abstract

This paper presents a new high-speed rail vehicle concept which allows curve negotiations at higher velocities. The proposed architecture introduces an actuated sliding carriage between the vehicle frame and bogie, allowing the supported frame to move laterally relative to the bogie, together with a separate tilt actuator that rotates the vehicle frame relative to the carriage. These two mechanisms are intended to control the position and orientation of the upper vehicle mass dynamically so that the rail-level resultant load path can be managed during curved-track operation.

A coupled lateral–roll model is developed and used for constrained time-domain simulation over an entry-transition curve. Passive, lateral-shift-only, tilt-only, and combined shift–tilt configurations are compared using the same route input. The study provides an initial dynamic assessment of the proposed architecture and shows how lateral mass placement and controlled frame roll contribute as distinct mechanisms for improved curve negotiation.

Keywords:
multibody dynamics, mechanical system design, centre-of-mass control, active suspension, feedforward control, dynamic stability, vehicle dynamics

1. Introduction

Carbody tilting is an established means of increasing railway curve speed while limiting the uncompensated lateral acceleration experienced by passengers. The Advanced Passenger Train (APT) combined electronic control, hydraulic tilt jacks, and a mechanical tilting bolster in one active-tilt system [1, 2, 3]. The associated control problem remains multi-objective: the commanded roll must follow the deterministic curvature and cant transition without responding excessively to track irregularities, while roll angle, roll rate, suspension travel, actuator capacity, and motion-sickness-related effects remain bounded [4, 5, 6]. Active hydraulic roll systems have further demonstrated that controlled carbody roll can be introduced while retaining a passive anti-roll load path when active operation is unavailable [7].

Lateral motion of the carbody relative to the bogie can be controlled by applying active forces across the secondary suspension. In several of these approaches, the aim is to keep the carbody near its centred position relative to the bogie, rather than command a deliberate lateral offset. Mellado et al. study active lateral suspension for conventional bogies, while Orvnäs et al. investigate it to improve ride comfort [8, 9]. A hold-off device addresses the same problem by limiting lateral suspension displacement to improve comfort, particularly during curve negotiation [10]. Alfi et al. study active air-spring control to improve ride quality and safety against crosswinds [11].

This coupling has motivated integrated lateral and roll controllers. Zhou et al. combined active tilt with active lateral secondary suspension in a high-speed railway vehicle and treated the interaction between the two actuator channels within a common vehicle model [12]. Di Gialleonardo et al. subsequently investigated coordinated control of an active lateral suspension and hydraulically actuated roll system, assessing increased curve speed against ride-quality, running-safety, and actuator-demand measures [13]. Consequently, the combination of lateral and roll actuation is already established in the literature. The principal objectives of these systems are the control of conventional carbody–bogie-relative motion, passenger comfort, suspension travel, and tilt compensation.

The integrated systems described in prior research control lateral and roll motion between the carbody and bogie. Here, a sliding carriage is inserted between the vehicle frame and bogie. Lateral actuation moves the carriage relative to the bogie, while the tilt actuator rotates the frame relative to the carriage. The added body therefore provides a controlled lateral translation of the upper structure in addition to frame tilt.

This combined approach is attractive because lateral shift and tilt influence the curve problem through different physical mechanisms. Lateral shifting changes the position of the mass centre directly. Tilting changes the lateral projection of the vehicle-frame mass and creates an inward roll action. Used together, the two motions may allow a new high-speed train concept to keep the rail-resultant offset between the tracks.

The vehicle is represented as a coupled lateral–roll multibody system comprising three principal bodies: the vehicle frame, sliding carriage, and bogie. Curve-negotiation feasibility is as-

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sessed primarily through the rail-level resultant line of action. If the resultant moves beyond the available rail-support width, the corresponding operating condition is considered to have exceeded the safety margin.

To assess the proposed architecture, four vehicle configurations are considered: a passive system, active tilt only, active lateral shift only, and combined lateral shift and tilt. A force-based dynamic formulation is used so that the commanded lateral displacement and roll motion arise from physical actuator forces and moments rather than prescribed vehicle kinematics. A feed-forward PD control strategy is then developed to coordinate the active degrees of freedom according to curve curvature, track cant, and vehicle speed. The resulting configurations are evaluated through transient curve-entry simulations and progressive speed sweeps to determine their influence on vehicle response, rail-resultant position, actuator demand, and the attainable rail-support margin.

The rail-resultant boundary used in this study is a preliminary overturning-type feasibility measure, not a complete railway running-safety criterion. Detailed wheel–rail contact, creep forces, wheel unloading, derailment ratio, structural gauging, passenger-comfort indices, actuator power, and fail-safe behaviour have not been assesed here. [14]. The paper examines whether lateral mass shifting, alone or combined with frame tilt, can increase the modelled curve-negotiation speed relative to the passive configuration while keeping the rail resultant within the defined gauge.

2. Vehicle Modelling

The vehicle is modelled as an end-view lateral–roll system with three coupled bodies: the vehicle frame, the sliding carriage, and the bogie. The arrangement is shown in Figure 1.

The generalised displacement vector is

$$\boldsymbol{q} = \begin{bmatrix} y_v & y_s & y_b & \theta_v & \theta_s & \theta_b \end{bmatrix}^T, \tag{1}$$

where y_v, y_s, and y_b are the lateral displacements of the vehicle frame, sliding carriage, and bogie, respectively. The corresponding roll coordinates are θ_v, θ_s, and θ_b. The wheelset lateral coordinate is y_w, the lateral track input is y_o, and the rail cant input is θ_o. For a vehicle speed V and local track curvature function represented as $\kappa(s)$, the quasi-static curve acceleration is

$$a_c(s) = V^2 \kappa(s), \tag{2}$$

which reduces to $a_c = V^2/R$ on the constant-radius part of the route.

The lateral actuator applies the physical force F_x between the sliding carriage and the bogie. Positive F_x acts in the positive y_s direction on the sliding carriage which points in the direction of the center of curvature. Because the actuator line of action is offset from the roll centres, the same force also produces roll moments on the sliding carriage and on the bogie.

The differential vertical actuator force produces the tilt moment. For this model, the lateral actuator is assumed to act in the same transverse plane as the lateral spring and damper connections between the sliding carriage and the bogie. Similarly, the differential tilt actuator is assumed to act through the same effective roll plane as the vertical/roll suspension connection between the vehicle frame and the sliding carriage.

2.1. Lateral Equations of Motion

Lateral equation for vehicle frame:

$$\begin{aligned} m_v \ddot{y}_v = {} & 2\left(K_{fy} y_s + C_{fy} \dot{y}_s\right) \\ & + 2(h_1 + h_2)\left(K_{fy} \theta_v + C_{fy} \dot{\theta}_v\right) \\ & - 2\left(K_{fy} y_v + C_{fy} \dot{y}_v\right) \\ & - 2h_2\left(K_{fy} \theta_s + C_{fy} \dot{\theta}_s\right) \\ & + m_v g \theta_o \\ & - m_v h_{g3} \ddot{\theta}_o \\ & - m_v a_c \\ & + S_{fy}. \end{aligned} \tag{3}$$

Lateral equation for sliding carriage:

$$\begin{aligned} m_s \ddot{y}_s = {} & 2\left(K_{sy} y_b + C_{sy} \dot{y}_b\right) \\ & + 2(h_2 + h_3)\left(K_{sy} \theta_s + C_{sy} \dot{\theta}_s\right) \\ & + 2h_4\left(K_{sy} \theta_b + C_{by} \dot{\theta}_b\right) \\ & + 2\left(K_{fy} y_v + C_{fy} \dot{y}_v\right) \\ & + 2h_2\left(K_{fy} \theta_s + C_{fy} \dot{\theta}_s\right) \\ & - 2(h_1 + h_2)\left(K_{fy} \theta_v + C_{fy} \dot{\theta}_v\right) \\ & - 2\left(K_{fy} y_s + C_{fy} \dot{y}_s\right) \\ & - 2\left(K_{sy} y_s + C_{sy} \dot{y}_s\right) \\ & + m_s g \theta_o \\ & - m_s h_{g2} \ddot{\theta}_o \\ & - m_s a_c \\ & + F_x \\ & + S_{sy} - S_{fy}. \end{aligned} \tag{4}$$

Lateral equation for bogie:

$$\begin{aligned} m_b \ddot{y}_b = {} & 2\left(K_{sy} y_s + C_{sy} \dot{y}_s\right) \\ & + 2h_5\left(K_{by} \theta_b + C_{by} \dot{\theta}_b\right) \\ & + 2\left(K_{by} y_w + C_{by} \dot{y}_w\right) \\ & - 2\left(K_{by} y_b + C_{by} \dot{y}_b\right) \\ & - 2\left(K_{sy} y_b + C_{sy} \dot{y}_b\right) \\ & - 2(h_2 + h_3)\left(K_{sy} \theta_s + C_{sy} \dot{\theta}_s\right) \\ & - 2h_4\left(K_{sy} \theta_b + C_{by} \dot{\theta}_b\right) \\ & + m_b g \theta_o \\ & - m_b h_{g1} \ddot{\theta}_o \\ & - m_b a_c \\ & - F_x \\ & - S_{sy}. \end{aligned} \tag{5}$$

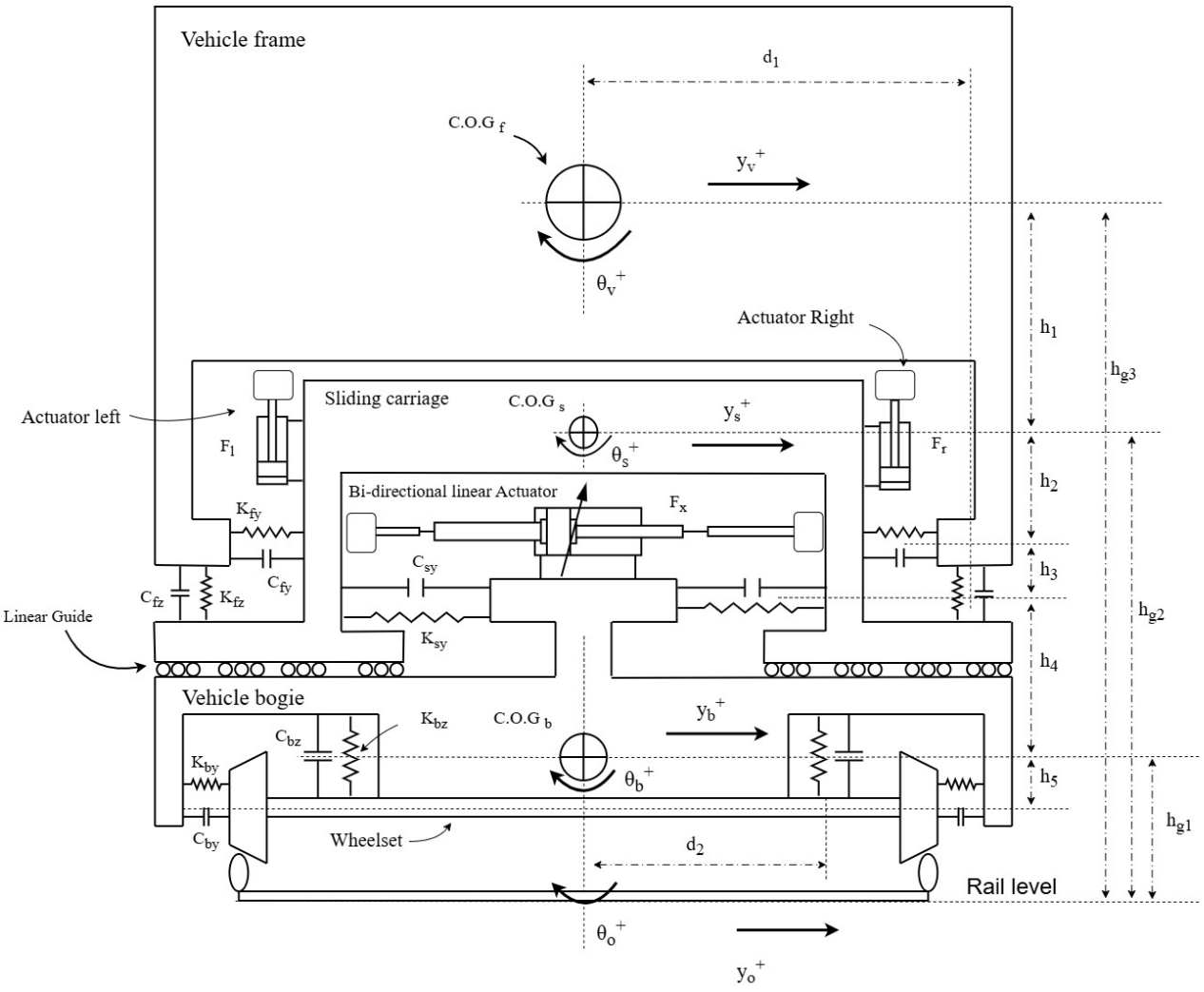


Figure 1: End-view schematic of the proposed vehicle architecture.

2.2. Roll Equations of Motion

Roll equation for vehicle frame:

$$\begin{aligned} I_{vr}\ddot{\theta}_v = {} & 2(h_1+h_2)\left(K_{fy}y_v + C_{fy}\dot{y}_v\right) + 2(h_1+h_2)h_2\left(K_{fy}\theta_s + C_{fy}\dot{\theta}_s\right) + 2d_1^2\left(K_{fz}\theta_s + C_{fz}\dot{\theta}_s\right) - 2(h_1+h_2)^2\left(K_{fy}\theta_v + C_{fy}\dot{\theta}_v\right) \\ & - 2(h_1+h_2)\left(K_{fy}y_s + C_{fy}\dot{y}_s\right) - 2d_1^2\left(K_{fz}\theta_v + C_{fz}\dot{\theta}_v\right) + m_v g\,(y_v - y_s) - I_{vr}\ddot{\theta}_o - (h_1+h_2)S_{fy} + M_t + S_{vs}. \end{aligned} \tag{6}$$

Roll equation for sliding carriage:

$$\begin{aligned} I_{sr}\ddot{\theta}_s = {} & 2h_2(h_1+h_2)\left(K_{fy}\theta_v + C_{fy}\dot{\theta}_v\right) + 2h_2\left(K_{fy}y_s + C_{fy}\dot{y}_s\right) + 2d_1^2\left(K_{fz}\theta_v + C_{fz}\dot{\theta}_v\right) + 2(h_2+h_3)\left(K_{sy}y_s + C_{sy}\dot{y}_s\right) \\ & - 2h_2\left(K_{fy}y_v + C_{fy}\dot{y}_v\right) - 2h_2^2\left(K_{fy}\theta_s + C_{fy}\dot{\theta}_s\right) - 2d_1^2\left(K_{fz}\theta_s + C_{fz}\dot{\theta}_s\right) - 2(h_2+h_3)\left(K_{sy}y_b + C_{sy}\dot{y}_b\right) \\ & - 2(h_2+h_3)^2\left(K_{sy}\theta_s + C_{sy}\dot{\theta}_s\right) - 2(h_2+h_3)h_4\left(K_{sy}\theta_b + C_{sy}\dot{\theta}_b\right) + m_s g\,(y_s - y_b) - I_{sr}\ddot{\theta}_o \\ & + 2d_1^2\left(K_{lgy}\theta_b + C_{lgy}\dot{\theta}_b\right) - 2d_1^2\left(K_{lgy}\theta_s + C_{lgy}\dot{\theta}_s\right) - (h_2+h_3)(F_x + S_{sy}) - M_t - S_{vs} + h_2 S_{fy} + S_{sb}. \end{aligned} \tag{7}$$

Roll equation for bogie:

$$\begin{aligned} I_{br}\ddot{\theta}_b = {} & 2h_4\left(K_{sy}y_s + C_{sy}\dot{y}_s\right) - 2h_5^2\left(K_{by}\theta_b + C_{by}\dot{\theta}_b\right) - 2h_5\left(K_{by}y_w + C_{by}\dot{y}_w\right) - 2d_2^2\left(K_{bz}\theta_b + C_{bz}\dot{\theta}_b\right) \\ & - 2h_4\left(K_{sy}y_b + C_{sy}\dot{y}_b\right) - 2h_4(h_2+h_3)\left(K_{sy}\theta_s + C_{sy}\dot{\theta}_s\right) - 2h_4^2\left(K_{sy}\theta_b + C_{sy}\dot{\theta}_b\right) + 2h_5\left(K_{by}y_b + C_{by}\dot{y}_b\right) \\ & + 2d_1^2\left(K_{lgy}\theta_s + C_{lgy}\dot{\theta}_s\right) - 2d_1^2\left(K_{lgy}\theta_b + C_{lgy}\dot{\theta}_b\right) - I_{br}\ddot{\theta}_o - h_4(F_x + S_{sy}) - S_{sb} + S_{b0}. \end{aligned} \tag{8}$$

2.3. Geometric Deflections and Mechanical Stops

The symbols S_{fy}, S_{sy}, S_{vs}, S_{sb}, and S_{b0} represent mechanical stop reactions associated with finite allowable relative motion between vehicle components. These limits represent physical constraints such as actuator stroke limits, guide clearances, suspension travel, and structural interference between adjacent bodies. The stop reactions are zero while the corresponding relative displacement or rotation remains within its prescribed range. Once a mechanical travel limit is exceeded, the associated stop is activated as a stiff spring–damper reaction.

The spring and guide deflections are not always equal to a simple coordinate difference such as $y_v - y_s$ or $y_s - y_b$. Each connection point is displaced by both the lateral translation of its body and the lateral motion produced by roll through its vertical lever arm. In this work, a *geometric deflection* means the relative lateral displacement measured at the actual spring, damper, or guide location after these roll-induced contributions have been included. This is the quantity that determines how much a mechanical connection is compressed or stretched.

For the vehicle-frame to sliding-carriage lateral connection, the geometric deflection is

$$\delta_{fy} = y_v - y_s - (h_1 + h_2)\theta_v + h_2\theta_s, \tag{9}$$

and for the sliding-carriage to bogie lateral guide it is

$$\delta_{sy} = y_s - y_b - (h_2 + h_3)\theta_s - h_4\theta_b. \tag{10}$$

The corresponding roll-stop coordinates are

$$\theta_{vs} = \theta_v - \theta_s, \qquad \theta_{sb} = \theta_s - \theta_b, \qquad \theta_{b0} = \theta_b - \theta_o. \tag{11}$$

The translational stop reaction for a generic lateral deflection δ with limit $\delta_{\max}$ is written as

$$\begin{aligned}&\mathcal{B}(\delta, \dot{\delta}; \delta_{\max}, K_{\text{stop}}, C_{\text{stop}}) \\ &= \begin{cases} 0, & |\delta| \le \delta_{\max}, \\ -K_{\text{stop}}(\delta - \delta_{\max}) - C_{\text{stop}} \max(\dot{\delta}, 0), & \delta > \delta_{\max}, \\ -K_{\text{stop}}(\delta + \delta_{\max}) - C_{\text{stop}} \min(\dot{\delta}, 0), & \delta < -\delta_{\max}. \end{cases}\end{aligned} \tag{12}$$

where $\mathcal{B}$ returns a lateral force. The corresponding rotational stop reaction is defined separately because it returns a moment rather than a force. The generic roll coordinate ϕ represents one of the relative roll coordinates

$$\phi \in \{\theta_{vs}, \theta_{sb}, \theta_{b0}\}. \tag{13}$$

For any ϕ in this set, with limit $\phi_{\max}$,

$$\begin{aligned}&\mathcal{B}_\theta(\phi, \dot{\phi}; \phi_{\max}, K_{\theta,\text{stop}}, C_{\theta,\text{stop}}) \\ &= \begin{cases} 0, & |\phi| \le \phi_{\max}, \\ -K_{\theta,\text{stop}}(\phi - \phi_{\max}) - C_{\theta,\text{stop}} \max(\dot{\phi}, 0), & \phi > \phi_{\max}, \\ -K_{\theta,\text{stop}}(\phi + \phi_{\max}) - C_{\theta,\text{stop}} \min(\dot{\phi}, 0), & \phi < -\phi_{\max}. \end{cases}\end{aligned} \tag{14}$$

$$\begin{aligned} S_{fy} &= \mathcal{B}\left(\delta_{fy}, \dot{\delta}_{fy}; \delta_{fy,\max}, K_{fy,\text{stop}}, C_{fy,\text{stop}}\right), \\ S_{sy} &= \mathcal{B}\left(\delta_{sy}, \dot{\delta}_{sy}; \delta_{sy,\max}, K_{sy,\text{stop}}, C_{sy,\text{stop}}\right), \\ S_{vs} &= \mathcal{B}_\theta\left(\theta_{vs}, \dot{\theta}_{vs}; \theta_{vs,\max}, K_{vs,\text{stop}}, C_{vs,\text{stop}}\right), \\ S_{sb} &= \mathcal{B}_\theta\left(\theta_{sb}, \dot{\theta}_{sb}; \theta_{sb,\max}, K_{sb,\text{stop}}, C_{sb,\text{stop}}\right), \\ S_{b0} &= \mathcal{B}_\theta\left(\theta_{b0}, \dot{\theta}_{b0}; \theta_{b0,\max}, K_{b0,\text{stop}}, C_{b0,\text{stop}}\right). \end{aligned} \tag{15}$$

2.4. Simulation Environment and Transition-Curve Input

The time-domain simulations are implemented in MATLAB using the force-balance equations above. The vehicle travels at a prescribed forward speed V, while the route position is advanced using

$$s(t) = Vt. \tag{16}$$

The simulated route is composed of an initial tangent, an entry clothoid, and a constant-radius circular curve. The entry clothoid is used so that curvature does not appear as a step input to the vehicle. Instead, curvature increases linearly with distance through the transition, which is the defining feature of a clothoid in railway alignment design. This choice follows established transition-curve practice and the related railway transition-curve literature [15, 16, 17].

Let s_E be the start of the transition, L_T the transition length, and $s_C = s_E + L_T$ the start of the circular curve. The normalised transition coordinate is

$$\xi(s) = \begin{cases} 0, & s < s_E, \\ \dfrac{s - s_E}{L_T}, & s_E \le s < s_C, \\ 1, & s \ge s_C. \end{cases} \tag{17}$$

The curvature profile used by the simulation is then

$$\kappa(s) = \kappa_{\max}\xi(s), \qquad \kappa_{\max} = \frac{1}{R}. \tag{18}$$

The rail cant is not ramped linearly. To avoid abrupt roll-rate changes at the ends of the transition, the cant angle uses the cubic smooth-step function

$$\begin{aligned} \chi(\xi) &= 3\xi^2 - 2\xi^3, \\ \theta_o(s) &= \theta_{o,\max}\chi(\xi), \\ \theta_{o,\max} &= \tan^{-1}\left(\frac{c_{\max}}{G}\right). \end{aligned} \tag{19}$$

This choice gives zero cant-rate at the beginning and end of the transition, while still reaching the specified full cant on the circular curve. The derivatives passed into the vehicle equations are obtained by the chain rule:

$$\begin{aligned} \dot{\theta}_o &= V\frac{\mathrm{d}\theta_o}{\mathrm{d}s}, \\ \ddot{\theta}_o &= V^2\frac{\mathrm{d}^2\theta_o}{\mathrm{d}s^2}, \\ \frac{\mathrm{d}\chi}{\mathrm{d}s} &= \frac{6\xi - 6\xi^2}{L_T}, \\ \frac{\mathrm{d}^2\chi}{\mathrm{d}s^2} &= \frac{6 - 12\xi}{L_T^2}. \end{aligned} \tag{20}$$

The same distance-domain profile is also used to form the track centreline for posture plots:

$$\begin{aligned} \psi(s) &= \psi(0) - \int_0^s \kappa(\sigma)\,\mathrm{d}\sigma, \\ X(s) &= \int_0^s \cos\psi(\sigma)\,\mathrm{d}\sigma, \\ Y(s) &= \int_0^s \sin\psi(\sigma)\,\mathrm{d}\sigma. \end{aligned} \tag{21}$$

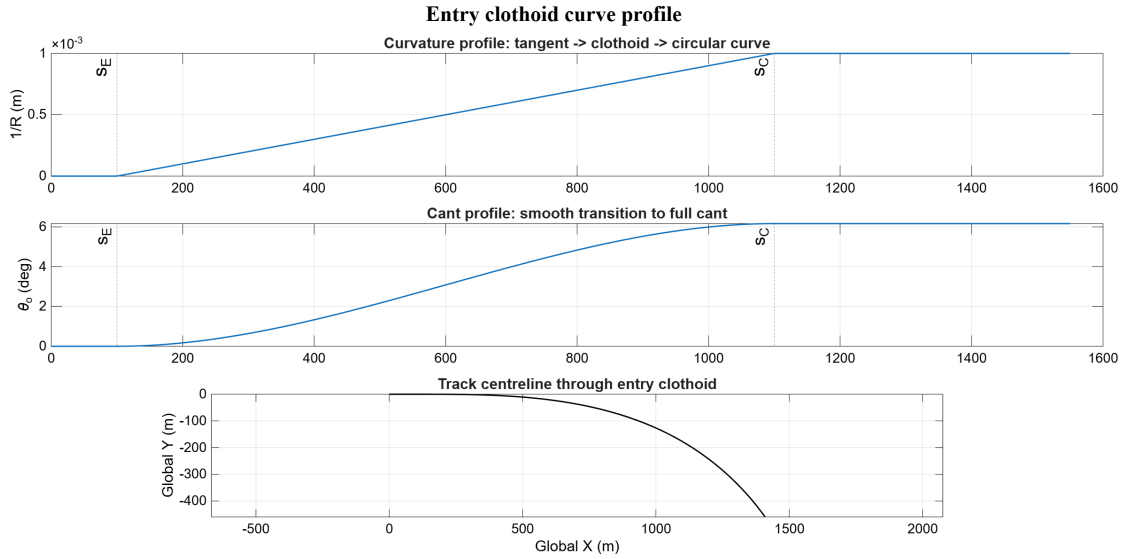


Figure 2: Simulation route definition used for the entry-transition studies. Curvature increases linearly through the clothoid, while rail cant follows a smooth transition before both reach their constant-radius values.

The negative sign in the heading equation follows the plotting convention used in this work, where positive lateral motion is inward toward the curve centre.

The effective rail-plane acceleration components used for the resultant-force and controller-target calculations are

$$\begin{aligned} a_z(s) &= g\cos\theta_o(s) + a_c(s)\sin\theta_o(s), \\ a_y(s) &= a_c(s)\cos\theta_o(s) - g\sin\theta_o(s), \end{aligned} \tag{22}$$

and the corresponding equivalent cant-deficiency angle is

$$\lambda_{\text{track}}(s) = \tan^{-1}\left(\frac{a_y(s)}{a_z(s)}\right). \tag{23}$$

This angle provides the geometric demand that the tilt and lateral-shift controllers attempt to reduce through inward frame roll and inward mass translation.

2.5. Wheel–Rail Input Filter and Actuator Interpretation

The wheel–rail lateral input is represented by the second-order filter

$$\ddot{y}_w + 2\zeta_w\omega_w\dot{y}_w + \omega_w^2 y_w = \omega_w^2 y_o. \tag{24}$$

In controlled time-domain simulations, the actuator commands are saturated and then passed through first-order actuator lags before being applied as the physical force and moment in the equations above:

$$\dot{F}_x = \frac{F_{x,\text{lim}} - F_x}{\tau_F}, \qquad \dot{M}_t = \frac{M_{t,\text{lim}} - M_t}{\tau_M}. \tag{25}$$

Here $F_{x,\text{lim}}$ and $M_{t,\text{lim}}$ are the saturated command values after the actuator force and moment limits have been applied. The first-order lag represents the finite actuator response bandwidth.

Although the wheelset lateral input filter is included in the model, it is not excited in the present simulations: the track-irregularity input is set to $y_o = 0$, so the wheelset lateral response remains $y_w = 0$. The present study therefore introduces the vehicle concept at a preliminary system-dynamics level. Detailed controller design and explicit wheel–rail contact dynamics, including the effects of track irregularities, are left for future work.

3. Control Strategy

The proposed control strategy uses the two actuator channels introduced in Section 2: the lateral force F_x and the roll moment M_t.

The controller is organised into three stages. First, the known route and operating condition are converted into shift and tilt reference targets. Second, the active feedback loops compare those targets with measured relative vehicle states and generate force or moment commands. Third, the commands are limited and passed through actuator dynamics before being applied to the vehicle equations.

3.1. Route-Based Reference Generation

The controller references are generated from the route geometry defined in Section 2.4. In the MATLAB simulation, the route position is $s = Vt$, so the local curvature, cant, and speed are known at each integration step. In a practical implementation, the same calculation could use stored route geometry or track information sensed ahead of the vehicle. Preview-based control has been studied for tilting trains, where advance track information allows the controller to anticipate an approaching curve [18]. The controller can therefore set a target from the upcoming curve condition rather than waiting for the vehicle response alone.

Using the curve acceleration, rail-plane acceleration components, and equivalent cant-deficiency angle already defined in Eqs. (2)–(23), the corresponding inward centre-of-mass correc-

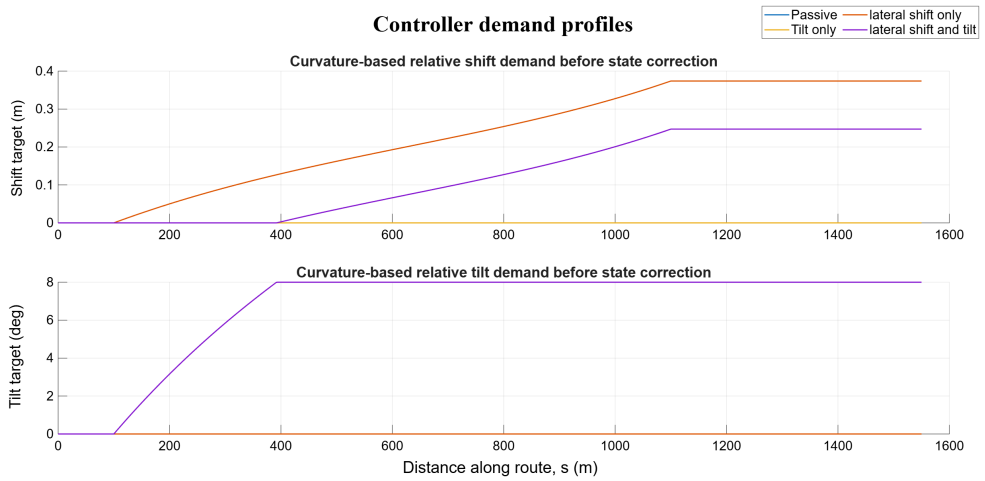


Figure 3: Route-based controller demand profiles before measured-state correction. The curves show the reference schedule generated from curvature, cant, and speed.

tion demanded by the local curve condition is written as

$$d_{\mathrm{COM}}(s) = H_{\mathrm{eff}} \tan(\lambda_{\mathrm{track}}(s)),$$
$$H_{\mathrm{eff}} = \frac{m_v h_{g3} + m_s h_{g2} + m_b h_{g1}}{m_v + m_s + m_b}. \tag{26}$$

For the numerical model used in the simulations, $H_{\mathrm{eff}} \approx 1.26$ m.

The COM correction is converted into actuator-coordinate targets using the geometric gains

$$G_y = \frac{m_v + m_s}{m_v + m_s + m_b}, \qquad G_\theta = \frac{m_v(h_1 + h_2)}{m_v + m_s + m_b}. \tag{27}$$

Here $G_y \approx 0.884$ maps carriage translation to whole-system COM translation, while $G_\theta \approx 0.802$ m rad^{-1} maps relative vehicle-frame tilt to the approximate COM movement associated with frame roll. The reference limits used in the controller are

$$\bar{y} = 0.6675 \text{ m}, \qquad \bar{\theta} = 8^\circ. \tag{28}$$

Before the measured-state correction is applied, the route-based targets are allocated according to the active control case. For shift-only operation, the geometric lateral target is

$$y_{\mathrm{tar}}^{(0)} = \mathrm{sat}_{\bar{y}}\left(\frac{d_{\mathrm{COM}}}{G_y}\right), \qquad \theta_{\mathrm{tar}}^{(0)} = 0. \tag{29}$$

For tilt-only operation,

$$\theta_{\mathrm{tar}}^{(0)} = \mathrm{sat}_{\bar{\theta}}\left(\frac{d_{\mathrm{COM}}}{G_\theta}\right), \qquad y_{\mathrm{tar}}^{(0)} = 0. \tag{30}$$

For combined operation, the tilt target is assigned first and the remaining COM correction is assigned to lateral shift:

$$\theta_{\mathrm{tar}}^{(0)} = \mathrm{sat}_{\bar{\theta}}\left(\frac{d_{\mathrm{COM}}}{G_\theta}\right), \qquad y_{\mathrm{tar}}^{(0)} = \mathrm{sat}_{\bar{y}}\left(\frac{d_{\mathrm{COM}} - G_\theta \theta_{\mathrm{tar}}^{(0)}}{G_y}\right). \tag{31}$$

The passive case sets both targets to zero.

Figure 3 shows the reference schedule created by this calculation. The lateral-shift-only case assigns the curve correction directly to carriage translation. The tilt-only case assigns the available correction to relative frame roll. In the combined case, the tilt target reaches its limit first, after which the remaining COM demand is passed to the lateral-shift target.

3.2. Measured-State Correction

During the time-domain simulation, the controller also reads the achieved relative states and the current mass-weighted COM position. The relative lateral-shift and tilt coordinates are

$$r_y = y_s - y_b, \qquad \dot{r}_y = \dot{y}_s - \dot{y}_b, \tag{32}$$

and

$$r_\theta = \theta_v - \theta_s, \qquad \dot{r}_\theta = \dot{\theta}_v - \dot{\theta}_s. \tag{33}$$

The actual inward COM position of the three-body model is

$$y_{\mathrm{COM,in}} = \frac{m_v y_v + m_s y_s + m_b y_b}{m_v + m_s + m_b}. \tag{34}$$

For lateral-shift-only operation, the state-corrected target is

$$y_{\mathrm{tar}} = \mathrm{sat}_{\bar{y}}\left[r_y + \frac{d_{\mathrm{COM}} - y_{\mathrm{COM,in}}}{G_y}\right]. \tag{35}$$

This form asks the carriage to move only by the remaining COM correction after the current vehicle displacement has already been accounted for. For tilt-only operation, the target remains the bounded curve-based relative-roll target:

$$\theta_{\mathrm{tar}} = \mathrm{sat}_{\bar{\theta}}\left[\frac{d_{\mathrm{COM}}}{G_\theta}\right]. \tag{36}$$

3.3. Lateral-Shift Controller

The lateral-shift controller is associated with the horizontal actuator between the sliding carriage and the bogie. Its purpose is to translate the supported upper structure relative to the bogie. A positive shift command corresponds to inward motion of the sliding carriage and vehicle frame toward the centre of curvature.

By increasing $y_s - y_b$ in the inward direction, the controller changes the mass-weighted centre-of-mass location.

For the supplied lateral-shift target y_{tar}, the error is

$$e_y = y_{\mathrm{tar}} - r_y. \tag{37}$$

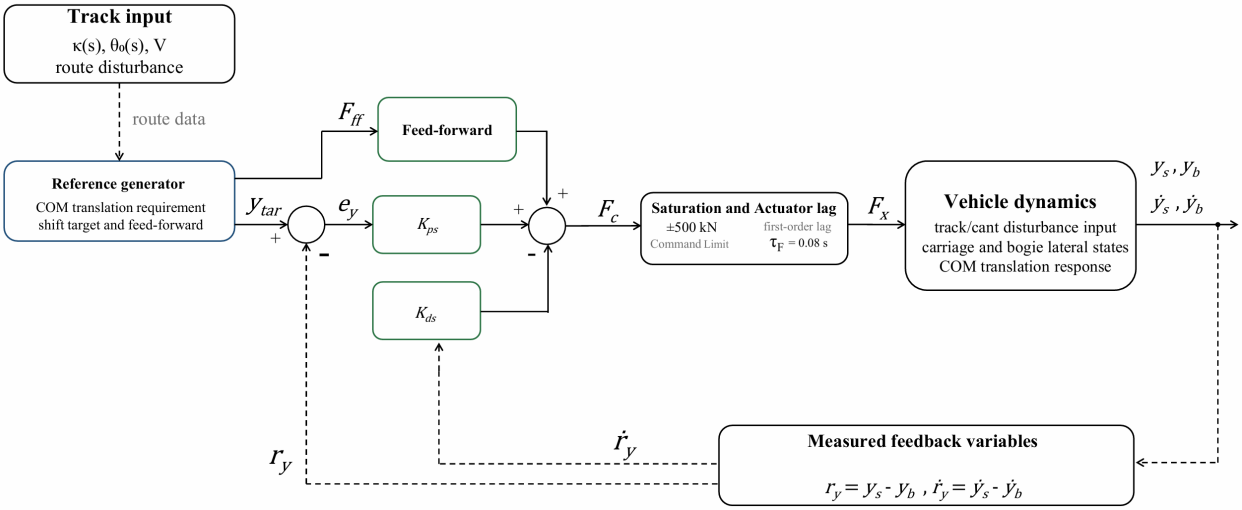


Figure 4: Closed-loop schematic for the lateral-shift controller.

If the achieved relative displacement r_y is smaller than the demanded inward shift, the error e_y is positive and the controller commands an inward actuator force on the sliding carriage. If the carriage overshoots the target, the sign of the error reverses and the commanded force is reduced or reversed. The command before actuator conditioning is

$$F_c = F_{\mathrm{ff}} + K_{ps} e_y - K_{ds} \dot{r}_y. \tag{38}$$

The feed-forward term F_{ff} is the anticipatory part of the lateral controller. It is estimated from the route-based shift target and the steady input–output gain of the vehicle plant, so it supplies the approximate force expected to hold the required relative shift under the local curve condition. The proportional term $K_{ps} e_y$ then acts like an active lateral stiffness between the demanded and achieved shift coordinates. The derivative term $-K_{ds}\dot{r}_y$ acts like active damping and resists rapid relative motion between the carriage and the bogie.

3.4. Tilt Controller

The tilt controller is associated with the differential actuator moment between the vehicle frame and the sliding carriage. Its controlled coordinate is the relative roll angle $\theta_v - \theta_s$.

The role of the tilt loop is therefore to create a controlled inward roll action of the vehicle frame. This changes the projected lateral position of the vehicle-frame mass and introduces an inward roll moment.

For the supplied tilt target θ_{tar}, the error is

$$e_\theta = \theta_{\mathrm{tar}} - r_\theta. \tag{39}$$

If the achieved relative roll r_θ is smaller than the demanded inward tilt, the error e_θ is positive and the actuator commands a positive moment on the vehicle frame. The equal-and-opposite reaction moment acts on the sliding carriage, so the controlled tilt motion is generated internally within the vehicle structure. The command before actuator conditioning is

$$M_c = M_{\mathrm{ff}} + K_{pt} e_\theta - K_{dt} \dot{r}_\theta. \tag{40}$$

The feed-forward term M_{ff} supplies the approximate steady roll moment expected from the route-based tilt target. The proportional term $K_{pt} e_\theta$ provides the active restoring moment needed to move the relative roll angle toward its target, while the derivative term $-K_{dt}\dot{r}_\theta$ damps relative roll-rate motion between the frame and the carriage as the demanded tilt changes with distance through the increasing curvature and cant.

3.5. Feed-Forward and Actuator Channel

The feed-forward terms are computed from the steady input–output gain of the vehicle plant. Let the relative output vector be

$$\boldsymbol{r} = \begin{bmatrix} r_y & r_\theta \end{bmatrix}^T, \qquad \boldsymbol{u} = \begin{bmatrix} F_x & M_t \end{bmatrix}^T. \tag{41}$$

To form this gain, the nonlinear vehicle equations are linearised about the passive straight-track equilibrium. With the generalised coordinates $\boldsymbol{q} = [y_v \; y_s \; y_b \; \theta_v \; \theta_s \; \theta_b]^T$ defined in Eq. (1), the vehicle-only state is

$$\begin{aligned} \boldsymbol{x}_v &= \begin{bmatrix} \boldsymbol{q}^T & \dot{\boldsymbol{q}}^T \end{bmatrix}^T, \\ \delta\dot{\boldsymbol{x}}_v &= \boldsymbol{A}_v \delta\boldsymbol{x}_v + \boldsymbol{B}_v \delta\boldsymbol{u}, \\ \delta\boldsymbol{u} &= \begin{bmatrix} \delta F_x & \delta M_t \end{bmatrix}^T. \end{aligned} \tag{42}$$

The wheel-filter states are excluded. The operating point has zero relative shift and tilt and zero actuator input, so the perturbations below coincide with the controller coordinates. The output map extracts the relative shift and tilt from the state perturbation:

$$\delta\boldsymbol{r} = \boldsymbol{C}_r \delta\boldsymbol{x}_v, \qquad \boldsymbol{C}_r = \begin{bmatrix} 0 & 1 & -1 & 0 & 0 & 0 & \boldsymbol{0}_{1\times 6} \\ 0 & 0 & 0 & 1 & -1 & 0 & \boldsymbol{0}_{1\times 6} \end{bmatrix}. \tag{43}$$

Accordingly, the steady gain from actuator input to relative output is written as

$$\boldsymbol{r}_{\mathrm{ss}} = \boldsymbol{G}_0 \boldsymbol{u}_{\mathrm{ss}}, \qquad \boldsymbol{G}_0 = -\boldsymbol{C}_r \boldsymbol{A}_v^{-1} \boldsymbol{B}_v. \tag{44}$$

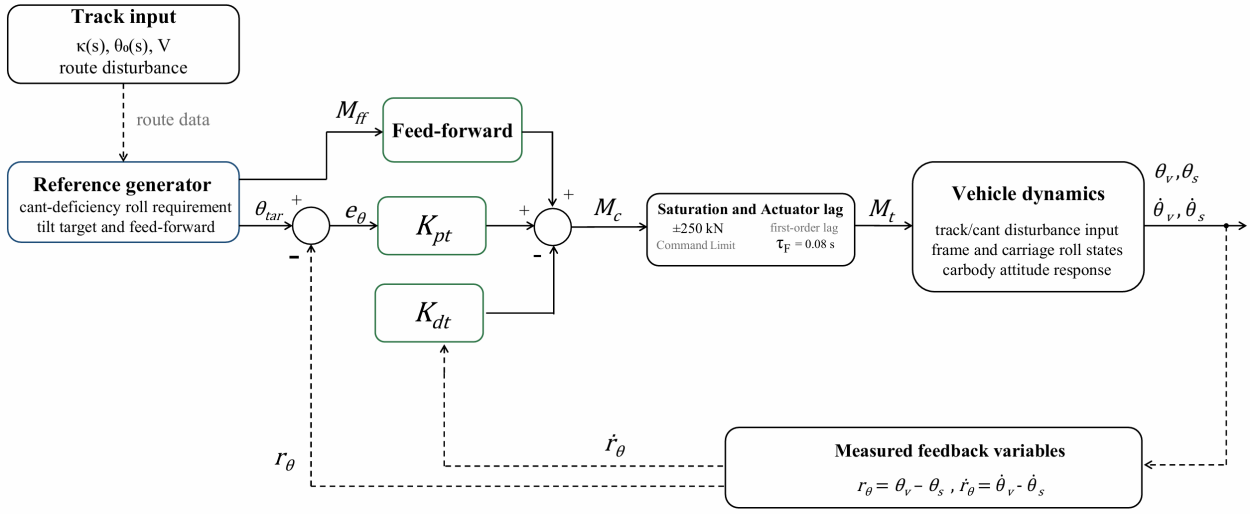


Figure 5: Closed-loop schematic for the vehicle-frame tilt controller.

The feed-forward input is then selected from this static inverse relation. For the combined controller,

$$\begin{bmatrix} F_{\mathrm{ff}} \\ M_{\mathrm{ff}} \end{bmatrix} = \boldsymbol{G}_0^{-1} \begin{bmatrix} y_{\mathrm{tar}} \\ \theta_{\mathrm{tar}} \end{bmatrix}. \tag{45}$$

For the single-actuator shift-only and tilt-only cases, the corresponding scalar part of $\boldsymbol{G}_0$ is used to find the feed-forward actuator command:

$$F_{\mathrm{ff}} = \frac{y_{\mathrm{tar}}}{G_{0,yy}}, \qquad M_{\mathrm{ff}} = \frac{\theta_{\mathrm{tar}}}{G_{0,\theta\theta}}. \tag{46}$$

The purpose of this feed-forward term is to preload the actuator command in the direction expected from the known curve profile. The feedback terms then correct the remaining error caused by transient dynamics, actuator lag, and mechanical stop interaction.

The controller command is not applied directly to the vehicle equations. The lateral force command is limited by

$$F_{x,\mathrm{lim}} = \mathrm{sat}_{500\ \mathrm{kN}}(F_c), \tag{47}$$

and the tilt moment command is limited by

$$M_{t,\mathrm{lim}} = \mathrm{sat}_{250\ \mathrm{kN\,m}}(M_c). \tag{48}$$

The saturated commands are then passed through the first-order actuator lag model in Eq. (25), with $\tau_F = \tau_M = 0.08$ s. Thus F_c and M_c are controller commands, while F_x and M_t are the physical inputs entering the equations of motion.

3.6. Combined Shift and Tilt Operation

When both actuators are enabled, the two feedback loops act on the same vehicle plant. The tilt controller receives θ_{tar} and commands M_t, while the lateral-shift controller receives y_{tar} and commands F_x.

In the combined case, tilt is allocated first and the remaining COM correction is assigned to lateral shift:

$$\begin{aligned} \theta_{\mathrm{tar}} &= \mathrm{sat}_{\bar{\theta}}\left[\frac{d_{\mathrm{COM}}}{G_\theta}\right], \\ y_{\mathrm{tar}} &= \mathrm{sat}_{\bar{y}}\left[r_y + \frac{d_{\mathrm{COM}} - y_{\mathrm{COM,in}}}{G_y} \right. \\ &\qquad \left. - \frac{G_\theta\,(\theta_{\mathrm{tar}} - r_\theta)}{G_y}\right]. \end{aligned} \tag{49}$$

Tilt generates the initial inward relative roll and changes the projected frame-COM position. The remaining correction is then provided by lateral shifting, which moves the supported mass further inward relative to the rail support.

4. Comparative Time-Domain Simulation Results

The vehicle model is evaluated using nonlinear time-domain simulations over the complete entry-clothoid route. Each run uses the same route input and a fixed forward speed. Positive lateral displacement is toward the centre of curvature. The four configurations differ in their enabled actuator channels: passive operation sets $F_x = M_t = 0$; shift-only enables F_x; tilt-only enables M_t; and the combined case enables both channels.

The principal safety margin is the rail-resultant gauge margin,

$$m_\Gamma(s) = \frac{G}{2} - \left|H_{\mathrm{eff}} \tan(\lambda_{\mathrm{track}}(s)) - y_{\mathrm{COM,in}}(s)\right|, \tag{50}$$

where the expression inside the absolute value is the rail-level lateral offset between the route-imposed resultant demand and the actual inward COM position. A positive value of m_Γ means that the resultant remains inside the rail-support width. A zero value means that the resultant has reached the allowable support boundary for the rail-level moment balance. The simulation also checks the lateral body-coordinate margin,

$$m_y(s) = \frac{G}{2} - \max(|y_v(s)|,\ |y_s(s)|,\ |y_b(s)|)\,, \tag{51}$$

which is used as an additional conservative geometric check. The main safety criterion is based on the rail-level resultant, but the body-coordinate margin prevents the simulation from accepting a case in which the mass-weighted COM remains acceptable while one individual body has moved beyond the rail-support width. This precaution is important because the vehicle frame, carriage, and bogie carry large masses; allowing any one of these bodies to act laterally beyond the rail envelope could introduce a large roll contribution even if the combined COM location appears acceptable.

The stop clearances were assigned according to the following configuration definitions:

- In all four cases, the frame–carriage lateral deflection was limited to $|\delta_{fy}| \le 0.10\,\mathrm{m}$, the carriage–bogie relative roll was limited to $|\theta_s - \theta_b| \le 2^\circ$, and the bogie roll relative to the imposed cant was limited to $|\theta_b - \theta_o| \le 0.15^\circ$.
- The passive baseline used $|\delta_{sy}| \le 0.10\,\mathrm{m}$ and $|\theta_v - \theta_s| \le 1^\circ$.
- The lateral-shift-only case used the active lateral travel allowance $|\delta_{sy}| \le 0.6675\,\mathrm{m}$ while keeping the relative-tilt stop at $|\theta_v - \theta_s| \le 2^\circ$.
- The tilt-only case retained the passive lateral guide clearance $|\delta_{sy}| \le 0.10\,\mathrm{m}$ and used the active tilt allowance $|\theta_v - \theta_s| \le 8^\circ$.
- The combined case used both active allowances: $|\delta_{sy}| \le 0.6675\,\mathrm{m}$ and $|\theta_v - \theta_s| \le 8^\circ$.
- Once a clearance was exceeded, the corresponding stop reaction was applied with K_{stop} and C_{stop} for lateral stops, and with $K_{\theta,\mathrm{stop}}$ and $C_{\theta,\mathrm{stop}}$ for roll stops.

4.1. Cross-Configuration Time-Domain Comparison

The four configurations are compared at their selected last-successful speed-sweep points: $300\,\mathrm{km\,h^{-1}}$ for passive operation, $370\,\mathrm{km\,h^{-1}}$ for lateral shift only, $310\,\mathrm{km\,h^{-1}}$ for tilt only, and $380\,\mathrm{km\,h^{-1}}$ for combined lateral shift and tilt. The figures therefore describe the response at each configuration's selected operating point. Responses are plotted against distance travelled, s, with s_E marking the start of the entry transition and s_C the start of the constant-radius section.

4.1.1. Safety and Constraint Quantities

Figure 6 compares the rail-resultant quantities, mechanical deflections, and vehicle-frame orientation. The plotted COM-balance error is

$$e(s) = H_{\mathrm{eff}} \tan(\lambda_{\mathrm{track}}(s)) - y_{\mathrm{COM,in}}(s). \tag{52}$$

The rail-resultant offset increases towards the outer support boundary as the vehicle progresses through the transition. Correspondingly, the gauge margin decreases from approximately 0.72 m on straight track to its smallest values near s_C. No crossing of the support boundary is visible at the plotted resolution. The passive margin approaches zero, whereas the active cases retain residual margins of a few centimetres, with the two shift-enabled cases retaining the larger values. These results indicate proximity to the modelled rail-support boundary at the selected speeds.

The carriage–bogie lateral spring/guide deflection, δ_{sy}, peaks at approximately 0.55 m for shift only and 0.52 m for combined operation, below the active stop-engagement clearance of 0.6675 m. Both subsequently settle near 0.42 m. The passive and tilt-only traces approach the negative 0.10 m guide clearance.

The frame–carriage relative roll, $\theta_v - \theta_s$, approaches 8° during entry in both tilt-enabled cases, before decreasing to approximately 4.7° for tilt only and 2.7° for combined operation. The passive and shift-only cases instead approach their negative relative-roll clearances. The carriage–bogie relative roll, $\theta_s - \theta_b$, reaches the vicinity of the ±2° stop-engagement thresholds during the entry transient and subsequently remains close to, or slightly beyond, the negative threshold.

The bogie–cant roll difference, $\theta_b - \theta_o$, requires separate attention. Its final value is approximately −0.8° to −0.9°, exceeding the stated 0.15° stop-engagement clearance in magnitude. In the finite-stiffness stop model, crossing this clearance activates a restoring reaction. The plotted response therefore indicates appreciable motion beyond stop engagement.

The plotted absolute vehicle-frame roll, $\theta_o + \theta_v$, approaches approximately 14° for tilt only, 12° for combined operation, 8° for passive operation, and 7° for shift only. At these operating points, the combined configuration therefore exhibits less final frame roll than tilt only, while also producing substantial inward translation.

4.1.2. Displacement and Roll Coordinates

Figure 7 shows the six generalised displacement and roll coordinates. Positive lateral displacement denotes motion towards the centre of curvature.

The passive vehicle frame initially moves outward to approximately $y_v = -0.10\,\mathrm{m}$ and then recovers towards the track centreline. Tilt-only operation produces an oscillatory inward excursion followed by a final displacement near 0.10 m. The shift-enabled cases produce substantially greater inward frame displacement: shift only peaks near 0.55 m and settles around 0.49 m, whereas combined operation peaks near 0.67 m and settles around 0.57 m.

The sliding-carriage coordinate, y_s, peaks near 0.58 m for shift only and 0.56 m for combined operation, with both settling just below 0.50 m. The combined carriage response develops later than the shift-only response, consistent with the initial allocation of correction to tilt. The larger combined-case frame displacement therefore does not correspond to a larger carriage translation. In the passive and tilt-only cases, the carriage remains slightly outward of the track centreline. Bogie displacement, y_b, is much smaller in every case, remaining on the millimetre scale and settling near 5 mm to 7 mm. The main lateral motion is consequently concentrated in the supported upper bodies.

The tilt-enabled cases exhibit early vehicle-frame roll peaks near 10°, followed by oscillations and final θ_v values of approximately 8° for tilt only and 6° for combined operation. The passive and shift-only cases initially develop negative frame

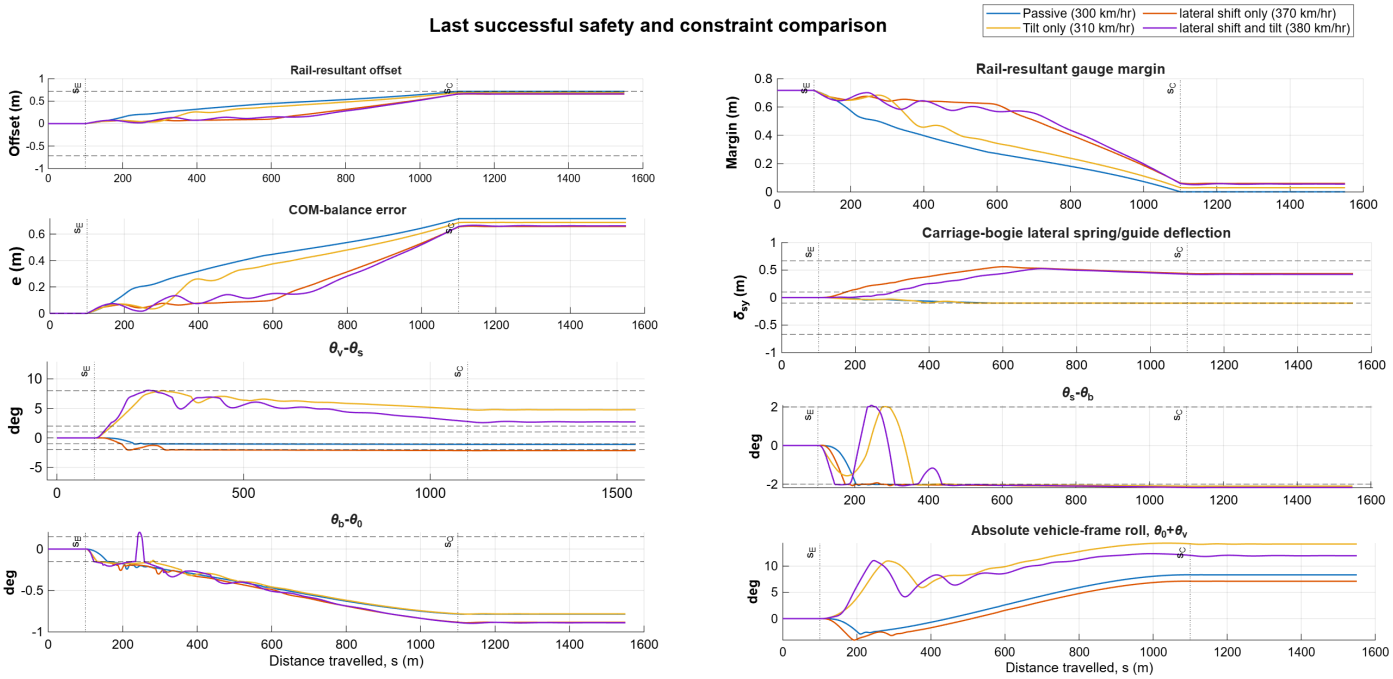


Figure 6: Comparison at the selected speed-sweep points: rail-resultant offset and gauge margin, COM-balance error, mechanical deflections, and absolute vehicle-frame roll.

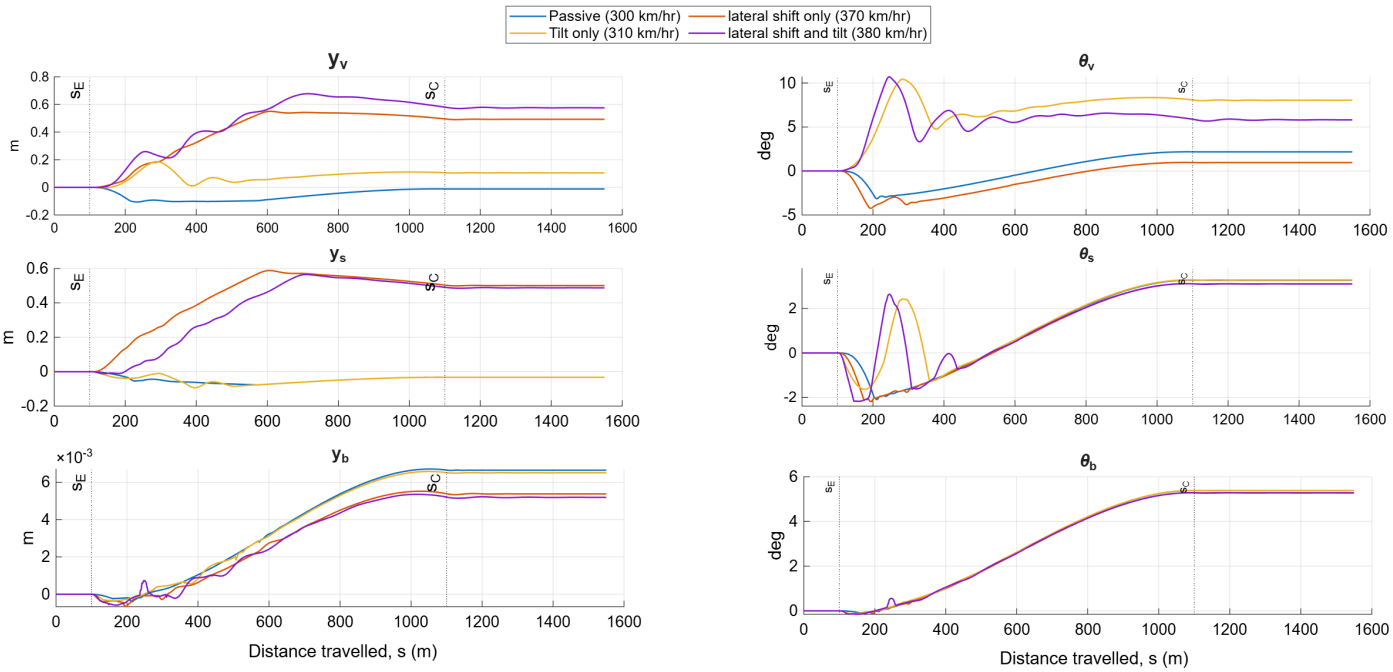


Figure 7: Generalised displacement and roll coordinates at the selected speed-sweep points.

roll before recovering to approximately 2° and 1°, respectively. The carriage roll, θ_s, also shows an initial negative excursion. In the tilt-enabled cases this is followed by a pronounced positive overshoot and further reversals, which are largely absent from the passive and shift-only traces. Later in the transition, all carriage-roll traces converge towards approximately 3°. The bogie-roll traces remain closely grouped and approach approximately 5.3°, although small entry transients distinguish the configurations. Beyond s_C, the coordinates undergo only small additional changes and approach approximately constant values.

4.1.3. Velocity and Roll-Rate Response

The corresponding lateral velocities and roll rates are compared in Fig. 8. The horizontal axis is distance travelled and the

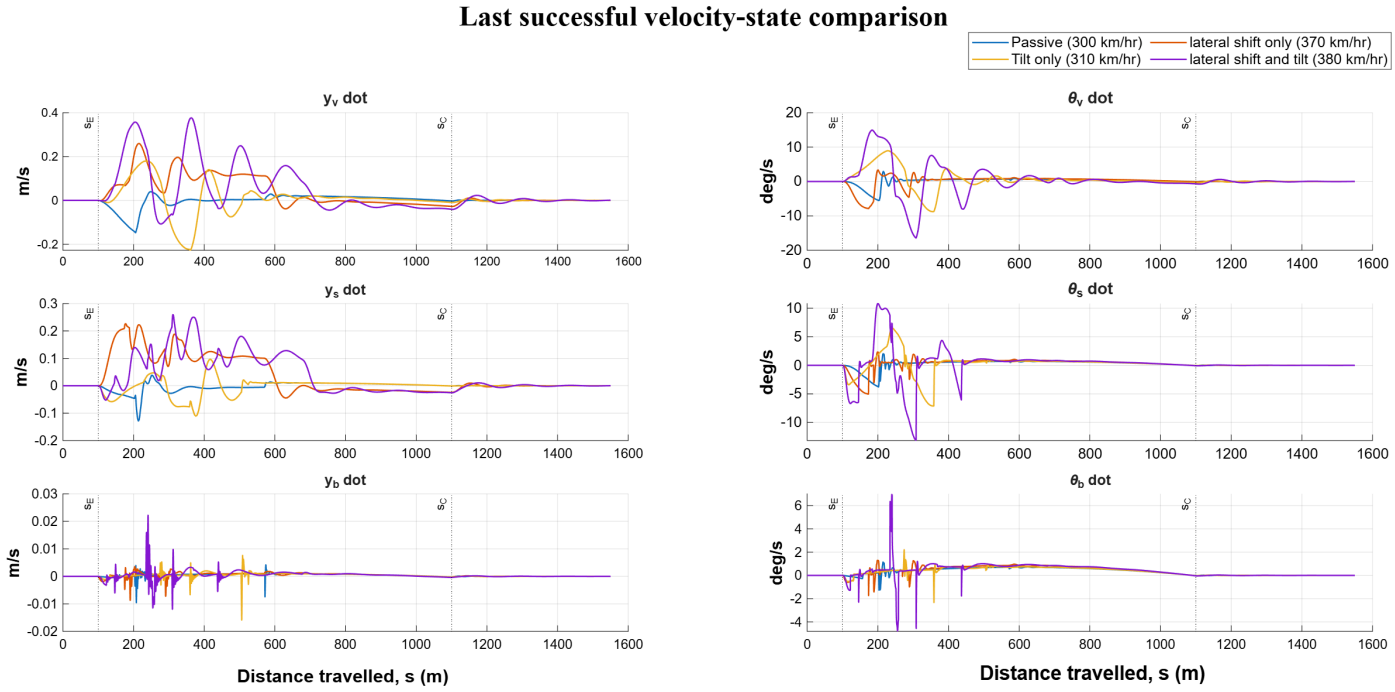


Figure 8: Lateral velocities and roll rates at the selected speed-sweep points, plotted against distance travelled.

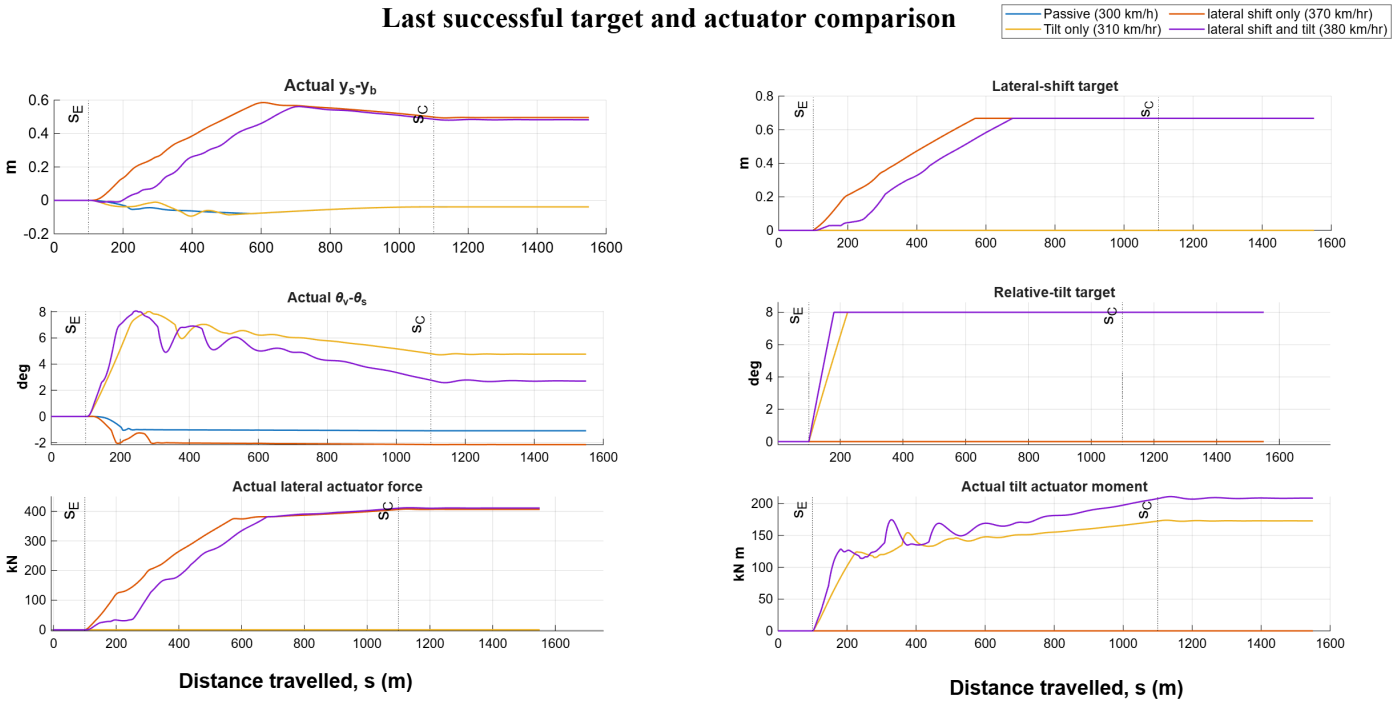


Figure 9: Relative shift and tilt references, achieved relative coordinates, and physical actuator force and moment at the selected speed-sweep points.

plotted quantities are time derivatives. The largest excursions occur during the early entry transition.

Combined operation produces the largest positive vehicle-frame lateral-velocity peaks, approximately $0.35\,\mathrm{m\,s^{-1}}$ to $0.38\,\mathrm{m\,s^{-1}}$, together with repeated direction reversals. The shift-only peak is approximately $0.25\,\mathrm{m\,s^{-1}}$ and its subsequent response is less oscillatory. Tilt-only operation also produces a significant reversal, with $\dot{y}_v$ reaching approximately $-0.22\,\mathrm{m\,s^{-1}}$ after its initial positive excursion. The passive frame initially moves outward before its velocity returns towards zero. The shift-enabled carriage velocities reach approximately $0.20\,\mathrm{m\,s^{-1}}$ to $0.26\,\mathrm{m\,s^{-1}}$; combined operation exhibits more repeated peaks than shift only. Bogie lateral velocities are smaller, although brief spikes remain visible during entry.

The combined vehicle-frame roll rate reaches approximately $+15\,^\circ\,s^{-1}$ and $-17\,^\circ\,s^{-1}$, compared with approximately $+9\,^\circ\,s^{-1}$ and $-9\,^\circ\,s^{-1}$ for tilt only. These reversals correspond to the early roll overshoot and oscillations in the displacement plots. The combined carriage roll rate also exhibits substantial excursions, reaching roughly $+10\,^\circ\,s^{-1}$ and $-13\,^\circ\,s^{-1}$. Bogie roll-rate traces contain short spikes, most prominently in the combined case, followed by a much smaller, slowly varying response through the remainder of the transition.

The oscillatory components diminish as the vehicle progresses along the route, and all plotted rates approach zero beyond s_C, apart from small settling transients. At its selected operating speed, combined operation therefore exhibits stronger transient lateral and roll motion than either single-actuator case. These state-rate plots do not directly quantify actuator stroke rates or establish passenger-comfort performance.

4.1.4. Controller Demand and Actuator Effort

Figure 9 compares the commanded references with the achieved relative coordinates and physical actuator outputs.

The lateral-shift reference reaches its 0.6675 m cap at approximately $s = 570$ m for shift only and $s = 680$ m for combined operation. The later increase in the combined shift demand is consistent with the allocation of early correction to tilt. The achieved relative shift, $y_s - y_b$, peaks near 0.58 m for shift only and 0.56 m for combined operation, before settling near 0.49 m and 0.48 m, respectively. Thus, both active shift channels retain a final reference-to-response difference of approximately 0.18 m to 0.19 m.

Both tilt references reach the 8° cap early in the transition, with the combined reference reaching the cap first. The achieved relative tilt initially approaches this value but subsequently decreases while the reference remains constant. Final relative tilts are approximately 4.7° for tilt only and 2.7° for combined operation, leaving residual tracking errors of approximately 3.3° and 5.3°. The capped reference therefore represents the requested correction, rather than the sustained tilt achieved by the vehicle.

The lateral actuator force rises to approximately 400 kN in both shift-enabled cases. The combined force builds more slowly during early entry, but the two traces converge later in the transition. The tilt actuator moment approaches approximately 170 kN m for tilt only and 210 kN m for combined operation, with more pronounced transient fluctuations in the combined trace. Disabled actuator channels remain at zero. The plotted physical outputs remain below the stated 500 kN force and 250 kN m moment limits, with no visible sustained operation at either limit. The persistent tracking errors therefore occur without sustained saturation of the plotted physical outputs; their causes cannot be attributed to saturation from these figures alone.

Within the reported sweep, lateral shift increases the selected last-successful speed from 300 km h^{-1} to 370 km h^{-1}, whereas tilt alone increases it to 310 km h^{-1}. Combined operation gives the highest selected value, 380 km h^{-1}, an additional 10 km h^{-1} beyond shift only. Lateral shifting therefore accounts for most of the increase in selected speed in this study. The additional increase with combined operation accompanies greater transient oscillation and a higher tilt-moment demand.

5. Conclusions and Future Work

This paper established a theoretical lateral–roll dynamic model for a new high-speed vehicle concept. The model includes the vehicle frame, sliding carriage, and bogie as coupled dynamic bodies, together with actuator lag, saturation limits, wheel–rail input filtering, and finite mechanical stops. The constrained time-domain simulations showed that the proposed active architecture can extend the modelled curve-running capability beyond the passive response.

The cross-configuration comparisons distinguish the two useful control effects. Lateral shifting provides direct inward mass placement by translating the supported structure relative to the bogie. Tilting provides an additional roll-based correction by rotating the elevated vehicle-frame mass and changing the gravity-inertia load line at rail level. The combined configuration gave the strongest response, supporting the main concept that curve negotiation can be improved by making vehicle mass placement and vehicle-frame roll active design variables.

The results should be interpreted as model-based evidence for the proposed concept. The achieved results depend on the assumed mass properties, suspension constants, actuator behaviour, guide travel, roll clearances, and mechanical stop definitions. Future work should therefore begin by replacing the assumed numerical constants with values derived from an actual vehicle layout and most importantly, introducing the criteria for passenger comfort. A sensitivity study should then identify which of these parameters most strongly controls rail-resultant margin, stop engagement, and actuator effort.

Further control development is also required. The present feed-forward-plus-feedback structure should be compared with gain-scheduled, state-feedback, robust, optimal, and predictive control strategies. The target-generation method should also be tested against different curve radii, cant profiles, transition lengths, and speed profiles so that the controller is evaluated over a wider operating envelope.

The allocation of corrective demand between tilt and lateral shift should also be investigated. The present tilt-first approach should be compared with shift-first allocation and simultaneous allocation using different weightings of the two mechanisms, assessing their effects on transient response.

Future work should extend the vehicle model to simulate the current design with higher-fidelity wheel–rail contact dynamics and track irregularities. Further model extensions should include creep-force modelling, wheel unloading, load transfer, actuator rate and power limits, fail-safe locking behaviour, and multi-car interaction.

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Appendix

This appendix collects the numerical parameters and case-definition variables used by the MATLAB model. The main text uses these quantities to define the vehicle equations, controller references, actuator channels, and mechanical travel limits, while the appendix keeps the detailed numerical list.

Model Variables and Numerical Values

Table 1 lists the principal variables used in the vehicle equations and in the MATLAB simulation environment. State variables are solved directly by the ODE model, while fixed parameter values correspond to the numerical model used for the dynamic curve studies.

Table 1: Vehicle-model and simulation-environment variables.

Variable	Value and description	Variable	Value and description
y_v	ODE state; lateral displacement of the vehicle frame.	θ_v	ODE state; roll angle of the vehicle frame.
y_s	ODE state; lateral displacement of the sliding carriage.	θ_s	ODE state; roll angle of the sliding carriage.
y_b	ODE state; lateral displacement of the bogie.	θ_b	ODE state; roll angle of the bogie.
y_w	0 in the simulations; wheelset lateral disturbance-filter output.	y_o	0; no lateral track-irregularity input.
θ_o	Route input; rail cant angle from Eq. (19).	a_c	$V^2\kappa(s)$; local curve acceleration.
V	Prescribed forward speed; held constant during each route simulation.	$s(t)$	Vt; distance travelled along the route.
ξ	0 to 1; normalized transition coordinate.	χ	$3\xi^2 - 2\xi^3$; smooth-step cant-shaping function.
X, Y	Integrated centreline coordinates used for route/posture plots.	ψ	Integrated track heading angle.
a_y, a_z	Rail-plane lateral and vertical acceleration components.	λ_{track}	Equivalent cant-deficiency angle used for target generation.
m_v	15 000 kg; vehicle-frame mass.	m_s	4000 kg; sliding-carriage mass.
m_b	2500 kg; bogie mass.	g	9.81 m s^{-2}; gravitational acceleration.
I_{vr}	25 000 kg m^2; vehicle-frame roll inertia.	I_{sr}	4500 kg m^2; sliding-carriage roll inertia.
I_{br}	1500 kg m^2; bogie roll inertia.	G	1.435 m; track gauge.
K_{fy}	5.0×10^6 N m^{-1} per side; frame–carriage lateral stiffness.	C_{fy}	2.5×10^5 N s m^{-1} per side; frame–carriage lateral damping.
K_{fz}	6.2×10^5 N m^{-1} per side; frame–carriage vertical/roll stiffness.	C_{fz}	3.3×10^4 N s m^{-1} per side; frame–carriage vertical/roll damping.
K_{sy}	2.6×10^5 N m^{-1} per side; carriage–bogie lateral-guide stiffness.	C_{sy}	3.3×10^4 N s m^{-1} per side; carriage–bogie lateral-guide damping.
K_{by}	3.5×10^7 N m^{-1} per side; bogie–wheelset lateral stiffness.	C_{by}	1.6×10^4 N s m^{-1} per side; bogie–wheelset lateral damping.
K_{bz}	2.0×10^6 N m^{-1} per side; bogie vertical/roll stiffness.	C_{bz}	2.0×10^4 N s m^{-1} per side; bogie vertical/roll damping.
K_{lgy}	3.086×10^5 N m^{-1} per side; carriage–bogie roll-guide stiffness.	C_{lgy}	1.543×10^4 N s m^{-1} per side; carriage–bogie roll-guide damping.
d_1	0.90 m; half-spacing of frame/carriage tilt springs.	d_2	1.00 m; half-spacing of bogie vertical springs.
h_1	0.90 m; frame lever above the sliding carriage.	h_2	0.25 m; carriage upper/lower lever term.
h_3	0.10 m; carriage-to-bogie guide lever addition.	h_4	0.45 m; bogie roll lever in carriage–bogie guide.
h_5	0.09 m; bogie–wheelset lateral spring roll lever.	h_{g1}	0.37 m; bogie COM height above rail plane.
h_{g2}	0.85 m; sliding-carriage COM height above rail plane.	h_{g3}	1.52 m; vehicle-frame COM height above rail plane.
F_x	Saturated at 500 kN; lateral actuator force between carriage and bogie.	M_t	Saturated at 250 kN m; differential tilt moment between frame and carriage.

Continued on next page

Variable	Value and description	Variable	Value and description
F_l, F_r	Left and right vertical actuator forces forming $M_t = d_1(F_r - F_l)$.	τ_F	0.08 s; lateral actuator first-order lag.
τ_M	0.08 s; tilt actuator first-order lag.	R	1000 m; circular-curve radius.
$\kappa_{\max}$	1.0×10^{-3} m^{-1}; full circular-curve curvature.	$c_{\max}$	0.155 m; maximum rail cant.
$\theta_{o,\max}$	0.1076 rad (6.16°); full cant angle.	s_E	100 m; start of entry clothoid after tangent.
L_T	1000 m; entry-clothoid transition length.	L_C	450 m; constant-radius circular-curve length.
s_{end}	1550 m; total simulated route length.	ω_w	31.42 rad s^{-1}; wheel–track input-filter natural frequency.
ζ_w	0.20; wheel–track input-filter damping ratio.		
δ_{fy}	Geometric frame–carriage lateral deflection from Eq. (9).	δ_{sy}	Geometric carriage–bogie lateral-guide deflection from Eq. (10).
S_{fy}, S_{sy}	Stop reaction forces for lateral relative motion.	S_{vs}, S_{sb}, S_{b0}	Stop reaction moments for relative roll motion.
K_{stop}	Stop stiffness used after a mechanical clearance is exceeded.	C_{stop}	Stop damping applied only while motion is moving farther into the stop.
$\delta_{fy,\max}$	Case dependent; frame–carriage lateral stop clearance.	$\delta_{sy,\max}$	Case dependent; carriage–bogie lateral stop clearance.
$\theta_{vs,\max}$	Case dependent; relative frame–carriage tilt limit.	$\theta_{sb,\max}$	2°; relative carriage–bogie roll limit.
$\theta_{b0,\max}$	0.15°; bogie roll limit relative to the rail cant.	$\boldsymbol{x}$	State vector used for first-order state-space assembly.
$\boldsymbol{u}$	$[F_x\ M_t]^T$; physical actuator input vector.		

Controller Gains and Feed-Forward Matrix

Table 2 lists the numerical controller constants used in the time-domain simulations. The proportional and derivative terms are applied to the relative lateral shift and relative roll coordinates, while the remaining constants define the target allocation and actuator conditioning used by the controller implementation.

Table 2: Controller gains and control-channel constants.

Variable	Value and description	Variable	Value and description
K_{ps}	3.5×10^5 N m^{-1}; proportional gain for the lateral-shift controller.	K_{ds}	5.0×10^4 N s m^{-1}; derivative gain for the lateral-shift controller.
K_{pt}	1.0×10^6 N m rad^{-1}; proportional gain for the tilt controller.	K_{dt}	1.2×10^5 N m s rad^{-1}; derivative gain for the tilt controller.
$F_{x,\max}$	500 kN; lateral actuator saturation limit.	$M_{t,\max}$	250 kN m; tilt actuator saturation limit.
τ_F	0.08 s; lateral actuator first-order lag.	τ_M	0.08 s; tilt actuator first-order lag.
G_y	0.8837; COM shift per unit carriage–bogie lateral shift.	G_θ	0.8023 m rad^{-1}; COM shift per unit relative frame tilt used in target allocation.
H_{eff}	1.2616 m; effective height of the total vehicle mass above the rail plane.	$\boldsymbol{G}_0^{-1}$	Static inverse gain used for feed-forward actuator preloading; given in Eq. (53).

The static feed-forward matrix used by the controller was

$$\boldsymbol{G}_0^{-1} = \begin{bmatrix} 4.9384 \times 10^5 & -1.1281 \times 10^5 \\ -1.6660 \times 10^4 & 7.6335 \times 10^5 \end{bmatrix}. \tag{53}$$

This matrix maps the target vector $[y_{\text{tar}},\ \theta_{\text{tar}}]^T$ to the feed-forward actuator vector $[F_{\text{ff}},\ M_{\text{ff}}]^T$, where y_{tar} is in metres and θ_{tar} is in radians.

Case-by-Case Simulation Variables

Table 3 lists the case-specific variables used to define the four dynamic simulations. The first column gives the MATLAB-style case variable or the corresponding model quantity. The vehicle equations remain the same in all cases; only actuator availability, controller reference availability, and mechanical stop clearances are changed.

Table 3: Case-by-case actuator and mechanical-limit variables.

Variable or setting	Passive baseline	Lateral shift only	Tilt only	Lateral shift and tilt
`caseDef.id`	`passive`	`shift_only`	`tilt_only`	`shift_tilt`
`caseDef.enableShift`	False; $F_x = 0$	True	False; $F_x = 0$	True
`caseDef.enableTilt`	False; $M_t = 0$	False; $M_t = 0$	True	True
`caseDef.lockShift`	False	False	False	False
Carriage–bogie lateral DOF	Dynamic	Dynamic, force driven	Dynamic, passive guide	Dynamic, force driven
y_{tar}	Not used	COM-error based target	Not used	Remaining COM-error target after tilt allocation
θ_{tar}	Not used	Not used	Curve-based relative-roll target	Curve-based relative-roll target allocated first
$F_{x,\max}$	Not used	±500 kN	Not used	±500 kN
$M_{t,\max}$	Not used	Not used	±250 kN m	±250 kN m
τ_F	Not used	0.08 s	Not used	0.08 s
τ_M	Not used	Not used	0.08 s	0.08 s
`limits.deltaFy`	0.10 m	0.10 m	0.10 m	0.10 m
`limits.deltaSy`	0.10 m	0.6675 m	0.10 m	0.6675 m
`limits.ysMinusYb`	0.10 m	0.6675 m	0.10 m	0.6675 m
`limits.thetaVMinusThetaS`	1°	2°	8°	8°
`limits.thetaSMinusThetaB`	2°	2°	2°	2°
$\theta_{b0,\max}$	0.15°	0.15°	0.15°	0.15°
$K_{fy,\mathrm{stop}}$	5.0×10^7 N m^{-1}	5.0×10^7 N m^{-1}	5.0×10^7 N m^{-1}	5.0×10^7 N m^{-1}
$C_{fy,\mathrm{stop}}$	1.0×10^6 N s m^{-1}	1.0×10^6 N s m^{-1}	1.0×10^6 N s m^{-1}	1.0×10^6 N s m^{-1}
$K_{sy,\mathrm{stop}}$	5.0×10^7 N m^{-1}	5.0×10^7 N m^{-1}	5.0×10^7 N m^{-1}	5.0×10^7 N m^{-1}
$C_{sy,\mathrm{stop}}$	1.0×10^6 N s m^{-1}	1.0×10^6 N s m^{-1}	1.0×10^6 N s m^{-1}	1.0×10^6 N s m^{-1}
$K_{\theta,\mathrm{stop}}$	5.0×10^7 N m rad^{-1}	5.0×10^7 N m rad^{-1}	5.0×10^7 N m rad^{-1}	5.0×10^7 N m rad^{-1}
$C_{\theta,\mathrm{stop}}$	1.0×10^6 N m s rad^{-1}	1.0×10^6 N m s rad^{-1}	1.0×10^6 N m s rad^{-1}	1.0×10^6 N m s rad^{-1}
`p.stop.maxStateAbs`	100	100	100	100

The passive and tilt-only cases retain the sliding-carriage lateral degree of freedom even though the lateral actuator is disabled. Therefore, the carriage can still move through suspension compliance and stop reactions; it is simply not actively forced toward a lateral-shift target. The lateral-shift-only and combined cases increase the allowable carriage–bogie lateral guide travel to the active shift limit, while the active-tilt cases increase the allowable frame–carriage relative roll to the active tilt limit.